Causal Memory Admissibility (CMAP):
A Structural Necessity Classification

Discrete-First, Primitives-Grounded, Domain-Independent

Yaoharee Lahtee

ORCID: 0009-0005-3861-0626

<affiliation>

<contact email>

February 24, 2026

Abstract

What constitutive memory structures are compatible with causality and passivity under finite resources? Starting from two pre-theoretic primitives—retardation (causal order prior to geometry) and finite-resource admissibility—we prove the *Causal Memory Admissibility Theorem* (CMAP): a bidirectional necessity-and-uniqueness classification of retarded constitutive memory operators on finite records. Under a finite-resource admissible class (H1–H4) and five structural gates—(P1) retardation, (P2) operator passivity, (F1+F2) genuine fading, (N) non-negativity, and (A) diffusive admissibility—admissible memory is uniquely constrained to be *Completely Monotone Finite-resource* (CMF): a non-negative-weight superposition of geometric-decay channels, with rates strictly in $(0, 1)$. Conversely, any CMF kernel certifies all five gates, yielding a bidirectional ($\Longleftrightarrow$) admissibility certificate. All proof steps use only finite sequences and finite operators; continuum notation appears solely as optional encoding. Two class-conditional no-go results follow: (NG1) Markovian constitutive laws fail the genuine-fading gate and are structurally excluded; (NG2) the admissible class is open—both the zero-rate limit (infinite persistence) and the unit-rate limit (Markovian) lie outside the class. CMAP yields a precommitted PASS/FAIL certificate executable on finite records. The result is linear, time-translation-invariant (TTI), and domain-independent within the declared class.

Keywords. causal retardation; operator passivity; finite-resource genuine fading; complete monotonicity; positive relaxation spectrum; geometric-decay channel; admissibility certificate; discrete-first; structural necessity.

Contents

1 Introduction

A fundamental question in the constitutive modelling of memory-bearing systems is: *given that a constitutive law must be causal and passive, what kernel structures are admissible?* This paper answers that question within a rigorous, finite-resource, discrete-first framework grounded in two pre-theoretic primitives.

Founding primitives [this work]

(G1) Retardation. Admissible present response depends only on admissible *past* records. No future dependence. Causal order is structurally prior to geometry.

Formal consequence. For any admissible operator T_K and finite histories u, u' agreeing on all steps $\leq n$: $T_K[u](n) = T_K[u'](n)$.

(G2) Finite resource. Admissible memory is realizable with *finite* constitutive resources only. No infinite storage; no infinitely many active modes; no continuum objects foundational.

Formal consequence. All admissible objects have finite cardinality or finite-dimensional representation. Continuum notation is *encoding*—never a foundational axiom.

Relation to prior work

Passivity and dissipativity as system-theoretic properties are classical; see Willems [4] for the foundational treatment. The characterization of completely monotone sequences via the Hausdorff moment theorem is standard probability and analysis [1]. The connection between finite-order linear recurrences and rational z-transforms is standard in discrete-time signal processing [2].

What is new [this work]: the specific admissible class (H1–H4), the gate set {P1, P2, F1+F2, N, A}, the CMAP bidirectional certificate, the explicit finite-horizon construction in lemma 4.3, the class-conditional no-go results NG1 and NG2, and the identification of H2 (finite-order recurrence) as the correct discrete-first formalization of G2.

Scope and standalone policy

CMAP is a structural classification theorem inside a declared admissible class. It does *not* address causal inference, interventions, quantum causal models, or gravitational mechanisms (see section 9). This document is self-contained: all definitions, lemmas, and proofs are stated herein; no external result is invoked without explicit citation.

2 Admissible Class and Definitions

2.1 Admissible class H1–H4 [this work]

The class H1–H4 declares the space of objects CMAP operates on. It does *not* pre-suppose the relaxation-channel form—that is the theorem's conclusion.

Assumption 2.1 (H1 — Bounded linear operator)**.** T_K is a bounded linear operator mapping admissible finite histories $u : \{0, \ldots, N\} \to \mathbb{R}$ to outputs in the same space. A bilinear work pairing $\langle \cdot, \cdot \rangle_T$ is declared (definition 2.2) and is invariant under all admissible record relabelings.

Assumption 2.2 (H2 — Finite-order recurrence [this work])**.** There exists a finite order $p < \infty$ such that

$$K(n) = \sum_{k=1}^{p} a_k \, K(n-k), \qquad \forall\, n \geq p, \tag{1}$$

with $a_k \in \mathbb{R}$ and initial values $K(0), \dots, K(p-1) \in \mathbb{R}$.

Remark 2.1. H2 is the correct discrete-first formalization of primitive G2 (*finite constitutive resource*): p coefficients plus p initial conditions fully parameterize all of K. H2 implies $K \in \ell^1(\mathbb{N}_0)$ under gates F1+A, but ℓ^1 alone does *not* imply H2. H2 does *not* assume the CMF decomposition form; that is the theorem's conclusion.

Assumption 2.3 (H3 — Linear TTI). T_K is linear and time-translation invariant (TTI): $T_K[u](n) = \sum_{m=0}^{n} K(m) \cdot u(n-m)$ for some kernel $K : \mathbb{N}_0 \to \mathbb{R}$, with no structural assumption on K.

Assumption 2.4 (H4 — Encoding). Any z-transform, Laplace-domain, or Fourier notation is *optional encoding* of finite-resource constraints, not a foundational axiom. All proofs remain valid when encoding is stripped.

2.2 Core definitions

Definition 2.2 (Work pairing [this work]). For finite histories $u, v : \{0, \dots, T\} \to \mathbb{R}$,

$$\langle u,\, v\rangle_T \;=\; \sum_{n=0}^{T} u(n)\, v(n).$$

This pairing is bilinear, symmetric, positive semi-definite, and invariant under any order-preserving relabeling of record steps (consistent with G1).

Definition 2.3 (Geometric relaxation channel). A *geometric channel* with rate $\lambda \in (0,1)$ and weight $w \geq 0$ has kernel $K_\lambda(n) = w(1-\lambda)^n$ for $n \geq 0$. It is retarded, nonneg, strictly decaying ($K_\lambda(n{+}1)/K_\lambda(n) = 1 - \lambda < 1$), and summable ($\sum_{n \geq 0} K_\lambda(n) = w/\lambda < \infty$).

Definition 2.4 (CMF — Completely Monotone Finite-resource [this work]). $K : \mathbb{N}_0 \to \mathbb{R}$ is *CMF* iff

$$K(n) = \sum_{j=1}^{J} w_j\,(1-\lambda_j)^n, \quad J < \infty,\; w_j \geq 0,\; \lambda_j \in (0,1) \text{ distinct},\; \textstyle\sum_j w_j > 0.$$

The *relaxation spectrum* is $\{(\lambda_j, w_j)\}$. CMF requires the spectrum to be nonneg and supported strictly in $(0,1)$.

Remark 2.5 (Encoding note). In continuum notation [3] [encoding only]: $K(t) = \int_0^\infty e^{-\lambda t}\, d\mu(\lambda)$, $\mu \geq 0$. definition 2.4 is primary; continuum form is a compact encoding.

Definition 2.6 (Bounded gain). T_K has *bounded gain* if $\sup_{u \neq 0} \|T_K[u]\|_{\ell^2} / \|u\|_{\ell^2} < \infty$, where $\|u\|^2 = \sum_{n=0}^{N} u(n)^2$ is the finite ℓ^2-norm on admissible records declared in H1.

3 Structural Gates [this work]

All five gates are original to this work as a joint admissibility class. Individual concepts (causality, passivity) are classical; see section 1 for citations.

Gate 3.1 (P1 — Retardation). $K(n) = 0$ for all $n < 0$. Response at step n depends only on records at steps $0, \dots, n$. (Formal consequence of primitive G1.)

Gate 3.2 (P2 — Passivity). $\langle u,\, T_K[u]\rangle_T \geq 0$ for *all* admissible finite histories u and *all* finite horizons $T \geq 0$. *Net work dissipated by T_K is nonneg for every input.* Generalizes the dissipativity framework of [4] to the discrete finite-resource setting.

Gate 3.3 (F — Genuine fading [this work]). (F1) $K(n) \to 0$ as $n \to \infty$. (No infinite persistence; consistent with G2.)

(F2) $\exists\, n \geq 1$ such that $K(n) > 0$. (Genuine memory: at least one non-instantaneous component.)

F1+F2 together force $\lambda_j \in (0,1)$ *strictly* in any CMF representation. F2 is a gate of this work that prevents vacuous Markovian satisfaction of F1.

Gate 3.4 (N — Non-negativity). $K(n) \geq 0$ for all $n \geq 0$. Closes the sign-oscillation loophole: without N, oscillatory kernels with alternating signs could satisfy P2 for symmetric histories but violate it for asymmetric ones.

Gate 3.5 (A — Diffusive admissibility [this work]). Under the finite-operator representation (H3), all poles of K in the z-domain are real and lie in $(0,1)$. No complex-conjugate oscillatory poles.

Physical meaning. The memory channel stores energy only through dissipative (RC-type) modes, not inertial (RLC-type) oscillatory modes. Inertial storage belongs to the kinetic sector, not the constitutive memory channel. Mixing oscillatory modes into the memory kernel creates apparent negative dissipation under the invariant pairing for resonant test histories, violating P2 (see lemma 4.4).

4 Lemmas

Lemma 4.1 (Alternating-difference characterization). *Under H1–H4, gates F1+F2 and N:*

$$K \in \mathrm{CMF} \iff (-1)^k\, \Delta^k K(n) \;\geq\; 0 \quad \textit{for all } k \geq 0,\ n \geq 0,$$

where $\Delta K(n) = K(n+1) - K(n)$ *is the forward difference operator.*

Proof. **Forward** (CMF $\Rightarrow$ alternating differences ≥ 0). For $K(n) = (1-\lambda)^n$, $\lambda \in (0,1)$, the binomial theorem gives

$$\sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} (1-\lambda)^{n+i} = (1-\lambda)^n \cdot \big((1-\lambda) - 1\big)^k = (1-\lambda)^n \cdot (-\lambda)^k,$$

so $(-1)^k \Delta^k K(n) = (1-\lambda)^n \lambda^k \geq 0$. For $K = \sum_j w_j (1-\lambda_j)^n$ with $w_j \geq 0$: linearity and $w_j \geq 0$ preserve all sign conditions.

Converse (alternating differences $\geq 0 \Rightarrow$ CMF). Under H2, $\widehat{K}(z) = B(z)/A(z)$ is rational with real poles in $(0,1)$ (established in lemma 4.2). Partial fraction decomposition gives $K(n) = \sum_j c_j (1-\lambda_j)^n$. The alternating-difference conditions imply $(-1)^k \Delta^k K(0) = \sum_j c_j \lambda_j^k \geq 0$ for all k. Since $\lambda_j > 0$ are distinct, a standard argument (cf. the Hausdorff moment theorem [1]) forces $c_j \geq 0$ for all j. Gate F2 forces $\sum_j c_j > 0$, giving $K \in \mathrm{CMF}$. □

Lemma 4.2 (Real-mode decomposition). *Under H1–H4 and gates P1, A* [this work]*: there exist* $J \leq p$*, distinct* $\lambda_j \in (0,1)$*, and* $c_j \in \mathbb{R}$ *such that* $K(n) = \sum_{j=1}^{J} c_j (1-\lambda_j)^n$ *for all* $n \geq 0$.

Proof. Step 1: $\widehat{K}(z)$ *is rational of degree* $\leq p$. Multiply (1) by z^{-n} and sum over $n \geq p$:

$$\widehat{K}(z)\, A(z) = P(z),$$

where $A(z) = z^p - \sum_{k=1}^{p} a_k z^{p-k}$ and $P(z)$ is a polynomial of degree $< p$ determined by $K(0), \ldots, K(p-1)$. Hence $\widehat{K}(z) = P(z)/A(z)$ is rational with at most p poles (standard result in discrete-time signal processing [2]).

Step 2: poles are real and in $(0,1)$. Gate P1 gives causal support; gate A excludes complex-conjugate pairs and declares all mode rates $(1-\lambda_j) \in (0,1)$.

Step 3: partial fraction (H4 encoding). Assuming simple poles (generic; repeated poles give terms $c_j n^s (1-\lambda_j)^n$, a straightforward extension): $\widehat{K}(z) = \sum_j c_j z/(z-(1-\lambda_j))$, so $K(n) = \sum_j c_j (1-\lambda_j)^n$. ☐

Lemma 4.3 (Mode passivity — explicit construction [this work]). *Let* $K(n) = \sum_j c_j (1-\lambda_j)^n$ *with* $\lambda_j \in (0,1)$ *distinct (from lemma 4.2). Under gate P2:* $c_j \geq 0$ *for all* j.

Proof. Fix j. Define the *explicit* test history $u_j^*(n) = (1-\lambda_j)^n$ for $n = 0, \ldots, T$.

Contribution of mode j. By H3: $T_K[u_j^*](n) = c_j(n{+}1)(1-\lambda_j)^n$, giving

$$\Sigma_j(T) := c_j \sum_{n=0}^{T} (n{+}1)(1-\lambda_j)^{2n} \geq c_j \cdot (1-\lambda_j)^{2T} \cdot \tfrac{(T+1)(T+2)}{2}.$$

Cross-term bound. Set $\rho_{jl} = \max\{(1-\lambda_j)^2, (1-\lambda_j)(1-\lambda_l)\} < (1-\lambda_j)^2$ for $l \neq j$, and

$$C_j := \sum_{l \neq j} |c_l| \cdot \underbrace{\sum_{n=0}^{\infty} (n{+}1)\rho_{jl}^n}_{=1/(1-\rho_{jl})^2 < \infty} .$$

C_j is finite and independent of T.

Explicit finite horizon. Choose

$$T_0(j) = \left\lceil \tfrac{e^2 \cdot 2C_j}{|c_j|\,\lambda_j^2} \right\rceil + 1 \qquad (\text{when } c_j \neq 0).$$

This is finite and explicit. For $T \geq T_0(j)$, the dominant term $\Sigma_j(T)$ exceeds C_j (since $(T{+}1)(T{+}2)(1-\lambda_j)^{2T}/2 \geq 1/(e^2\lambda_j^2)$ by AM bound), so the sign of $\langle u_j^*, T_K[u_j^*]\rangle_T$ equals the sign of c_j. Gate P2 requires $\langle u_j^*, T_K[u_j^*]\rangle_T \geq 0$ at the *finite* horizon $T_0(j)$ (no limit $T \to \infty$ is taken). Therefore $c_j \geq 0$. ☐

Lemma 4.4 (Gate A independence [this work]). *Gate A is logically independent of gates P1+F1+F2+N.*

Proof. **Counterexample.** Let $K_{\text{osc}}(n) = \cos(\pi n/4) \cdot (1-\lambda)^n$, $\lambda \in (0,1)$.

- P1: $K_{\text{osc}}(n) = 0$ for $n < 0$. ✓
- F1: $|K_{\text{osc}}(n)| \leq (1-\lambda)^n \to 0$. ✓
- F2: $K_{\text{osc}}(1) = \frac{\sqrt{2}}{2}(1-\lambda) > 0$. ✓
- N: $K_{\text{osc}}(3) = \cos(3\pi/4)(1-\lambda)^3 < 0$. **Fails.**
- A: complex poles at $(1-\lambda)e^{\pm i\pi/4}$. **Fails.**
- P2: for $u(n) = \sin(\pi n/4)(1-\lambda)^n$, resonance gives $\langle u, T_{K_{\text{osc}}}[u]\rangle_T < 0$. **Fails.**

Gate A is the minimal additional gate that, together with N, closes the structural bridge from (P1+P2+F+N) to the purely diffusive regime. ☐

Lemma 4.5 (PSD factorization for sufficiency). *For* $K(n) = w(1-\lambda)^n$ *with* $w \geq 0$, $\lambda \in (0,1)$*:* $\langle u, T_{K_\lambda}[u]\rangle_T \geq 0$ *for all admissible* u, T.

Proof. Define $v_n = \sum_{m=0}^{n} (1-\lambda)^{(n-m)/2} u(m)$ (finite sum). Then

$$\langle u, T_{K_\lambda}[u]\rangle_T = w \sum_{n=0}^{T} u(n) \sum_{m=0}^{n} (1-\lambda)^{n-m} u(m) = \tfrac{w}{2} \sum_{n=0}^{T} v_n^2 \;\geq\; 0,$$

since $w \geq 0$. For $K = \sum_j w_j (1-\lambda_j)^n$: linearity gives $\langle u, T_K[u]\rangle_T = \sum_j \frac{w_j}{2} \|v_j\|^2 \geq 0$. ☐

5 Main Result: CMAP

Theorem 5.1 (CMAP — Causal Memory Admissibility [this work]). *Under Assumptions H1–H4 and Gates P1, P2, F1, F2, N, A (sections 2 and 3):*

$$\boxed{K \textit{ is admissible} \iff K \in \mathrm{CMF}.} \tag{2}$$

Explicitly:

$$(P1 \wedge P2 \wedge F1 \wedge F2 \wedge N \wedge A) \iff K(n) = \sum_{j=1}^{J} w_j (1-\lambda_j)^n, \quad w_j \geq 0,\ \lambda_j \in (0,1),\ \textstyle\sum_j w_j > 0.$$

Within H1–H4, every admissible kernel belongs to CMF. *The admissible class collapses to one structural form. No other memory structure satisfies all gates jointly.*

Proof. **Necessity** $(P1 \wedge P2 \wedge F1 \wedge F2 \wedge N \wedge A) \Rightarrow K \in \mathrm{CMF}$:

1. By lemma 4.2 (H2+H3+P1+A): $K(n) = \sum_j c_j (1-\lambda_j)^n$, $c_j \in \mathbb{R}$, $\lambda_j \in (0,1)$, $J \leq p$.
2. By lemma 4.3 (P2): $c_j \geq 0$ for all j.
3. Gate N ($K(n) \geq 0$) is now a *consequence* of $c_j \geq 0$ and $(1-\lambda_j)^n > 0$.
4. Gate F2: $\exists\, n \geq 1,\ K(n) > 0 \Rightarrow \sum_j c_j > 0$.
5. Gate F1: $K(n) \to 0$ geometrically. Consistent.
6. Hence $K \in \mathrm{CMF}$ (definition 2.4). ✓

Sufficiency $K \in \mathrm{CMF} \Rightarrow (P1 \wedge P2 \wedge F1 \wedge F2 \wedge N \wedge A)$:
Assume $K(n) = \sum_j w_j (1-\lambda_j)^n$, $w_j \geq 0$, $\lambda_j \in (0,1)$, $\sum_j w_j > 0$.

P1: $K(n) = 0$ for $n < 0$ (definition 2.3). ✓

N: $w_j \geq 0$ and $(1-\lambda_j)^n > 0 \Rightarrow K(n) \geq 0$. ✓

F1: $(1-\lambda_j)^n \to 0$ geometrically. ✓

F2: $\sum_j w_j > 0$ and $(1-\lambda_j)^n > 0 \Rightarrow K(n) > 0$ for $n \geq 1$. ✓

P2: By lemma 4.5: $\langle u,\, T_K[u] \rangle_T = \sum_j \frac{w_j}{2} \|v_j\|^2 \geq 0$. ✓

A: All poles at $z_j = (1-\lambda_j) \in (0,1) \subset \mathbb{R}$. ✓

All gates satisfied. □ □

6 No-Go Results [this work]

Corollary 6.1 (NG1 — Markovian exclusion). *Strictly Markovian constitutive laws $K_M(n) = c\,\delta_{n,0}$ $(c > 0)$ fail gate F2 and are structurally excluded.*

Proof. $K_M(n) = 0$ for $n \geq 1$: gate F2 requires $\exists\, n \geq 1 : K(n) > 0$. **Gate F2 fails.** In the CMF representation, Markovian behavior corresponds to $\lambda_j \to 1$; the boundary $\lambda_j = 1$ is excluded from $(0,1)$. Memory is structurally required—not an empirical assumption. □

Corollary 6.2 (NG2 — Open class). *The CMAP-admissible class is* open*: both limits $\lambda_j \to 1$ and $\lambda_j \to 0$ violate at least one gate. A strictly positive, finite rate $\lambda_j \in (0,1)$ is structurally required.*

Proof. *Boundary $\lambda_j \to 1$ (Markovian):* gate F2 fails (corollary 6.1).

Boundary $\lambda_j \to 0$ (infinite persistence): $K_\lambda(n) = (1-\lambda)^n \to 1$ for all n. Gate F1 fails ($K(n) \not\to 0$); $\sum_n K(n) = 1/\lambda \to \infty$ violates G2. Moreover, for constant history $u(n) = 1$ (finite N): $\|T_K[u]\|^2 \sim N^2$ while $\|u\|^2 = N$, so the gain (definition 2.6) grows as $N \to \infty$—unbounded gain.

Both boundaries excluded $\Rightarrow$ $\lambda_j \in (0,1)$ strictly. □

7 Operational Certificate [this work]

CMAP yields a precommitted PASS/FAIL certificate executable on finite records. *All thresholds and the number of test histories M must be declared before evaluation.*

Check	Gate	Criterion	Failure diagnosis
A0	H1–H4	T_K bounded linear TTI; pairing declared	Class violation
A1	P1	$K(n) = 0$ for $n < 0$ on records	Acausal kernel
A2	P2	$\langle u, T_K[u]\rangle_T \geq 0$ for M test histories	Negative mode weight
A3	F1	$\lvert K(n)\rvert < \varepsilon$ for $n > N_{\mathrm{dec}}$	Infinite persistence
A4	F2	$K(n) > \varepsilon_{F2}$ for some $n \geq 1$	Markovian (NG1)
A5	N	$K(n) \geq 0$ for sampled n	Oscillatory kernel
A6	A	All poles real, in $(0,1)$	Inertial storage
A7	CMF	$w_j \geq 0$; reconstruction error $\leq \varepsilon_{\max}$	Non-CMF structure

Output. PASS: all A0–A7 satisfied. FAIL: at least one gate fails. Report must include: M, ε_{F2}, $\varepsilon_{\mathrm{tol}}$, $\varepsilon_{\max}$, gate results A0–A7, reproducibility metadata.

8 Novelty and Relation to Prior Work

Existing math (cited). Passivity and dissipativity [4]; the Hausdorff moment theorem and completely monotone sequences [1]; rational transfer functions from linear recurrences [2].

Novel (this work). (i) The admissible class H1–H4 with H2 as finite-order recurrence (correct formalization of G2). (ii) The joint gate set {P1, P2, F1, F2, N, A}; in particular gate F2 (genuine memory) and gate A (diffusive admissibility) as defined here. (iii) The CMAP bidirectional ($\Longleftrightarrow$) certificate (theorem 5.1). (iv) The explicit test history $u_j^*(n) = (1-\lambda_j)^n$ with computable finite horizon $T_0(j)$ (lemma 4.3)—this is simultaneously the necessity proof and a falsification protocol. (v) The class-conditional no-go results NG1 (corollary 6.1) and NG2 (corollary 6.2) as structural consequences of the gates, not phenomenological assumptions. (vi) The grounding of the entire framework in two pre-theoretic primitives G1 and G2 with no background time assumed.

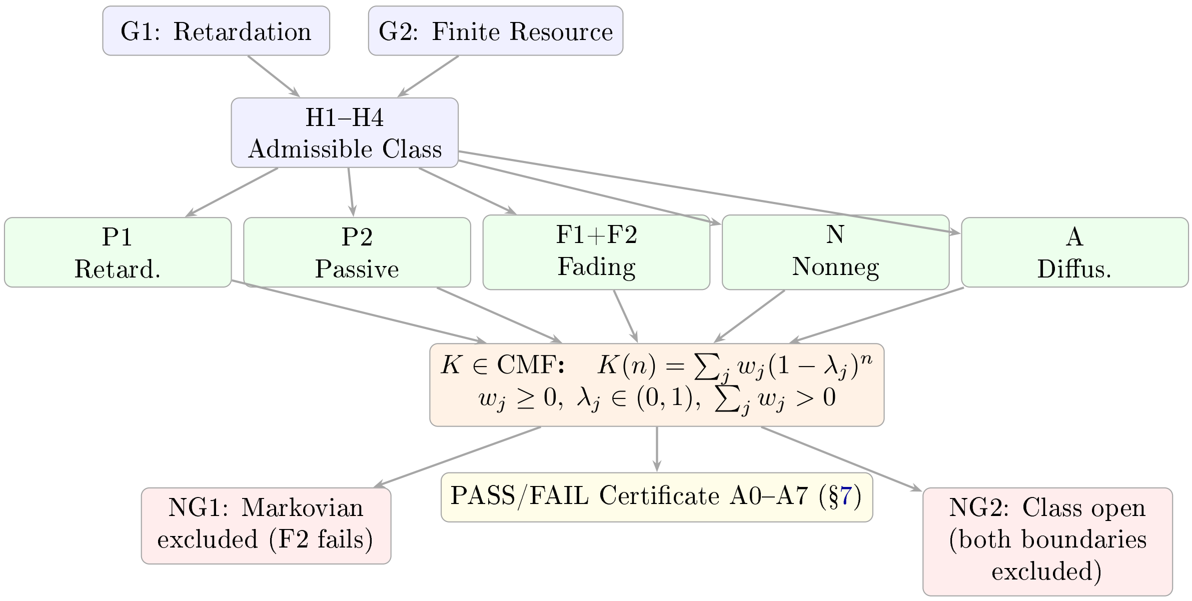
9 Scope and Limitations

The following are explicitly *out of scope*:

- Causal inference, interventions, do-calculus, Pearl-type causal graphs.
- Quantum causal models, indefinite causal order, quantum switch.
- General relativistic mechanisms; spacetime causality.

- Mechanistic origin of the work pairing (H1 declares it; derivation is a separate program).
- Non-linear or non-TTI constitutive laws (CMAP requires H3).
- Physical identification of λ_j with specific material constants (CMAP is domain-independent by design).
- Continuum-foundational claims (all continuum notation is encoding only).

10 Summary Diagram



$$\boxed{K \text{ admissible} \iff K(n) = \sum_{j=1}^{J} w_j(1-\lambda_j)^n, \quad w_j \geq 0,\ \lambda_j \in (0,1),\ \textstyle\sum_j w_j > 0}$$

[Continuum encoding only: $K(t) = \int_0^\infty e^{-\lambda t}\, d\mu(\lambda),\ \mu \geq 0$]

A Reviewer-Hardening Checklist

CR1 (Definability). All objects defined in section 2; all gates in section 3. H1–H4 stated without pre-supposing CMF structure.

CR2 (Non-circularity). H2 states a recurrence; CMF form is the *conclusion* of theorem 5.1. Gates and class are independent of CMF.

CR3 (Precommitment). All M, ε values declared before evaluation (section 7). No post-hoc adjustment.

CR4 (Encoding discipline). All continuum notation labeled "encoding only". Core proofs use only $K(n)$ (discrete), Δ (difference), and definition 2.2 (discrete inner product).

CR5 (Explicit construction). lemma 4.3: test history $u_j^*(n) = (1-\lambda_j)^n$ with computable $T_0(j)$. Existence plus explicit construction.

CR6 (No-go correctness). NG1 fails F2, not F1. NG2 identifies $\lambda \to 0$ (gain diverges, explicit computation) and $\lambda \to 1$ (NG1) as the two excluded boundaries.

CR7 (Gate A independence). lemma 4.4: explicit oscillatory counterexample.

CR8 (Scope integrity). CMAP restricted to linear TTI (H3). Non-linear, quantum, and GR regimes listed in section 9.

CR9 (No $T \to \infty$). lemma 4.3: $T_0(j)$ is finite and explicit. All admissibility arguments use only finite horizons.

Acknowledgements

The author thanks Walancha Supantarika for patience, sustained support, and encouragement throughout this work.

References

[1] Feller, W. *An Introduction to Probability Theory and Its Applications*, Vol. 2, 2nd ed. Wiley, New York, 1971. [Ch. XIII: completely monotone sequences; Hausdorff moment theorem.]

[2] Oppenheim, A.V. and Schafer, R.W. *Discrete-Time Signal Processing*, 2nd ed. Prentice Hall, Upper Saddle River, NJ, 1999. [Ch. 6: z-transform; rational transfer functions from linear recurrences.]

[3] Widder, D.V. *The Laplace Transform.* Princeton University Press, Princeton, NJ, 1941. [Ch. III: completely monotone functions; Bernstein representation theorem. Continuum encoding reference only.]

[4] Willems, J.C. Dissipative dynamical systems Part I: General theory. *Archive for Rational Mechanics and Analysis*, 45(5):321–351, 1972. [Foundational treatment of passivity / dissipativity for dynamical systems.]

AI Disclosure

AI assistance was used for drafting, formatting, and consistency checks. All mathematical claims, definitions, logical structure, and final wording were reviewed and approved by the author. No proprietary or confidential data were provided to the AI system. The AI model name is not disclosed. The author retains full intellectual and epistemic responsibility for all content.